

OpenCV-Based Microscope Automated Focusing & 3D Fusion System — User Manual

Software link:

https://drive.google.com/file/d/16daq7fprnsxukdZ4S2WjzEJeXOwqW67/view?usp=drive_link

1. System Environment & Boot Mechanism

- **Hardware Setup:** Ensure that your microscope camera (Type-C/USB Microscope) is properly connected to the computer.
- **Environment Bootstrapping:** The script includes a built-in virtual environment redirection feature. If the program is accidentally started using the system's global Python interpreter, it will **automatically detect and switch to the .venv virtual environment** located in the project directory—no manual path configuration required.
- **Output Location:** All generated PNG images, 360° interactive web pages (HTML), and 3D models are automatically saved into the `result/` folder in the project's root directory for easy access.

2. Phase 1: Main Menu — Input Source & Parameter Settings

Upon launching the program, the system enters a fullscreen **[Input Configurations]** window.

(Tip: You can press Esc at any time to toggle fullscreen off, and press Q outside of text inputs to quit the program completely).

2.1 Mode Selection (Choose One)

1. **Live CAM Mode:**
 - Used for real-time operations connected to a physical microscope camera.
 - The system automatically scans available camera indexes (compatible with DirectShow and MSMF backends).
 - Click "Start" to open the camera window, then **press Space** to begin automated recording and frame sampling based on your settings. The UI will display a live countdown timer and the number of captured frames.
2. **Select Recorded Video Mode:**
 - Click the "Browse" button to select a pre-recorded multi-focal video file from your computer (supports .mp4, .avi, .mov, etc.).
 - The system automatically parses the file and displays the precise duration on the right side of the UI.

2.2 Core Parameter Configuration

- **Recording Duration (Live CAM):** Set the total length of time to record from the live camera (Default: 6 seconds).
- **Sampling Interval:** Set the time step for extracting frames (Default: 0.3 seconds). This determines the density of focal plane layers sampled.
- **Output Filename:** Type a custom base name for your saved files (no need to type .png). The system automatically filters out illegal characters and prevents accidental file overwriting by incrementing

a suffix (e.g., _1, _2).

2.3 Background Automated Sampling & Extended Depth of Field (EDoF)

Clicking the "Start" button initiates an automated data pipeline while a loading progress bar is displayed:

1. **Quality Filtering:** Evaluates the clarity of all extracted frames using the Laplacian Variance algorithm, automatically discarding blurry frames (drops the bottom 45%) and removing redundant adjacent frames via average pixel differences.
2. **Rigid Alignment:** Utilizes the **ECC (Enhanced Correlation Coefficient) Algorithm** for sub-pixel image registration, eliminating ghosting and mechanical vibrations caused by manual focusing or hand tremors.
3. **Pixel-Level Fusion:** Calculates the post-Gaussian Laplacian response pixel by pixel across all aligned layers, **extracting only the sharpest pixels** to construct an all-in-focus composite image (stacked.png), solving the microscope's shallow depth of field issue.
4. **Refinement:** Automatically applies bilateral filtering (denoising while preserving edge details) and Gaussian-weighted sharpening to optimize image quality.

3. Phase 2: ROI (Region of Interest) Selection

Once the fusion is complete, the standard OpenCV interactive interface appears:

- **How to Operate:** **Click and drag your mouse** across the all-in-focus image to draw a bounding box around the micro-organism or target object you want to isolate and reconstruct.
- **Confirm/Cancel:** Press **Enter** to confirm the crop. The system will slice the multi-frame layer data within this region. Press **Esc** to cancel and safely return to the main menu.

4. Phase 3: Interactive Matting Editor (Core Workspace)

In this window, the removed (transparent) areas are overlaid onto a white background for live previewing. The editor provides a hybrid workflow of "Chroma-key Click" and "Smart Brush":

4.1 Hotkeys & Operations Quick Sheet

Action / Hotkey	Description
Left Click	Color Selection: Automatically removes regions with colors similar to the clicked pixel (BGR tolerance defaults to 35).
Right Click	Undo Color: Removes the last added color-removal rule, restoring that specific color mask.
B Key	Toggle Brush Mode: Turns the circular restoration brush ON/OFF to paint over and restore accidentally erased areas.
+ / - Keys	Adjust Brush Size: Increases or decreases the brush radius (Range: 5 to 100)

	pixels).
U Key	Multi-Level Undo: Steps backward through history snapshots (supports up to 30 steps of undo).
3 Key	Instant 360° Web Preview: Asynchronously generates a 360° interactive HTML page and opens it instantly in your web browser.
T Key	Cloud 3D Generation: Exports the current cutout and invokes the Tripo API to generate a textured, real-world PLY 3D model.
S Key	Save & Proceed: Asynchronously saves the precise transparent-background RGBA PNG image and advances to the next phase.
Q or Esc	Discard & Exit: Closes the editor without saving and returns safely to the main menu.

5. Phase 4: Output & Interactive Panel

After saving your image with S, a fullscreen **[Save Successful]** dashboard pops up, displaying four guiding choices:

1. **View 2D Image:**
 - Opens the newly created transparent foreground PNG file using your operating system's default photo viewer.
2. **360° Interactive View:**
 - **Priority Mechanism:** If multi-frame focal plane history slices are available, it packs them into sequential depth layers inside an HTML page. Otherwise, it generates a single-image orbital spin web page.
 - Automatically opens your browser so you can use your mouse to spin, rotate, and examine the depth strata of the micro-structure.
3. **Back to Main Menu (Next Video):**
 - Resets the application pipeline state and returns to the home screen to process the next sample.
4. **Quit Program:**
 - Safely shuts down all active OpenCV windows, closes the Tkinter runtime loop, releases system memory, and exits the process.